Complex Analysis and Differential Geometry

Question: 5 (Marks: 1)

The magnitude of $\exp(2+3i)$ is _____

Choices:

- $\exp(2)$
- $\exp(3)$
- $\exp(2*3)$
- $\exp(2/3)$

Virtulian Bird

Watermarkly

Complex Analysis and Differential Geometry

Question: 3 (Marks: 1)

$$f(z)=\frac{1}{z^2+1}$$

The function is defined everywhere in the finite plane except at the points

_____.

Choices:

- $z=\pm 1$
- $z=\pm 2$
- $z=\pm i$
- $z=\pm 2i$

Virtulian Bird

Watermarkly

Question: 3 (Marks: 1)

If $z = e^{i\theta}$, then $z^n =$ _____

Choices:

$z = \pm 1$

$z = \pm 2$

$z = \pm i$

$z = \pm 2i$

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Question: 4 (Marks: 1)

If $z = e^{i\theta}$, then $z^n =$ _____

Choices:

- $e^{i\theta}$
- $e^{-i\theta}$
- $e^{in\theta}$
- $e^{-in\theta}$

Virtulian Bird

Watermarkly

Complex Analysis and Differential Geometry

Question: 2 (Marks: 1)

The absolute value of the number, $z = 1-i$ is ____

Choices:

- 2
- $\sqrt{2}$
- 0
- 1

Virtulian Bird
Watermarkly

Question: 9 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $V_x =$ ____

Choices:

- -1
- 1
- 0
- 2

Virtulian Bird

Watermarkly

Question: 8 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $U_x =$ _____

Choices:

-1

1

0

2

Virtulian Bird

Watermarkly

Question: 7 (Marks: 1)

Let $f(z)=\frac{1}{z}$ is discontinuous at $z=$ _____

Choices:

- 0
- 1
- -1
- i

Virtulian Bird

TIME LEFT

59

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Complex Analysis and Differential Geometry

Question: 6 (Marks: 1)

Square of $-2-5i$ is _____

Choices:

- $-21+20i$
- $-2-5i$
- $-20-10i$
- $20+10i$

Virtulian Bird

Watermarkly

Question: 9 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $\int_C z =$ _____

Choices:

- -1
- 1
- 0
- 2

Virtulian Bird

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Complex Analysis and Differential Geometry

Question: 10 (Marks: 1)

The Polar Cauchy–Riemann equations on a pair of real-valued functions of two real variables r and v are _____.

Choices:

- $rU_r = V_\theta$ and $U_\theta = -rV_r$
- $U_r = V_\theta$ and $U_\theta = -V_r$
- $rU_\theta = V_r$ and $U_r = -rV_\theta$
- $U_r = V_\theta$ and $U_\theta = V_r$

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Question: 11 (Marks: 1)

In the complex valued function $f(z) = \frac{1}{re^{i\theta}}$, the value of $U_r =$ _____

Choices:

- $\frac{1}{r}$
- $\frac{\cos\theta}{r}$
- $\cos\theta$
- e^{θ}

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Question: 11 (Marks: 1)

In the complex valued function $f(z)=\frac{1}{re^{i\theta}}$, the value of $U_r =$ _____

Choices:

- $\frac{1}{r}$
- $\frac{\cos\theta}{r}$
- $\cos\theta$
- $e^{i\theta}$

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Complex Analysis and Differential Geometry

Question: 15 (Marks: 1)

Let $f(z)=2z^3+2z-7$, then $f'(z)$ is _____.

Choices:

- $6z^2+2$ correct
- z^2+2
- $2z^2-7$
- $2z^3+2$

Virtulian Bird

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Watermarkly

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Question: 14 (Marks: 1)

Let $f(z) = 3z^2 + 5z + 10$, then $f'(z)$ is _____

Choices:

- $5z$
- $6z$
- $6z - 5$
- $6z + 5$

correct

Virtulian Bird
Watermarkly

TIME LEFT
58

Question: 13 (Marks: 1)

The limit of $f(z)=z^2+3z-1$ at $z=i$ is _____

Choices:

- $3i$
- $3i-2$
- $2i-3$
- $2i-3$

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Complex Analysis

Question: 12 (Marks: 1)

The limit of $f(z) = z^2 + z + 1$ at $z = i - 1$ is _____.

Choices:

- 1
- -1
- i
- $-i$

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Complex Analysis and Differential Geometry

Question: 28 (Marks: 3)

Use Cauchy-Riemann equations to show that $f'(z)$ does not exist at any point if $f(z) = z - \bar{z}$, where $z = x + iy$

Answer:

Virtulian Bird

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Question: 24 (Marks: 1)

If $z = a + ib$ is a complex number, then _____.

Choices:

- $Re(z) = \frac{z + \bar{z}}{2}$
- $Re(z) = \frac{z - \bar{z}}{2}$
- $Re(z) = \frac{z.\bar{z}}{2}$
- $Re(z) = z + \bar{z}$

$= z + \bar{z}$ correct

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Watermarkly

Complex Analysis and Differential Geometry

Question: 20 (Marks: 1)

Let $z = 1 + i$, then exponential form of z is _____

Choices:

- $z = \sqrt{3} e^{\frac{\pi i}{4}}$
- $z = e^{\frac{\pi i}{4}}$
- $z = \sqrt{2} e^{\frac{\pi i}{3}}$
- $z = \sqrt{2} e^{\frac{\pi i}{4}}$

$z = \sqrt{2} e^{\frac{\pi}{4} i}$

correct

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Complex Analysis and

Question: 16 (Marks: 1)

Let $f(z) = 2z^2 - 7z + 1$, then $f'(-2i)$ is _____.

Choices:

- $-2i+1$
- $-4i-7$
- $-8i-7$
- $-8i+1$

correct

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Watermarkly

Complex Analysis and Differential Geometry

Question: 29 (Marks: 3)

Express the complex number $-\sqrt{3}+i$ in exponential form.

Answer:

Virtulian Bird

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Question: 25 (Marks: 1)

Which of the following are important part(s) of a function?

Choices:

- domain and range
- domain and rule
- range and rule
- rule only

correct

Virtulian Bird

Watermarkly

Question: 21 (Marks: 1)

The function $f(z) = \frac{1}{3z+i}$ is discontinuous at $z =$ _____

Choices:

- $3i$
- $-3i$
- $\frac{i}{3}$
- $-\frac{i}{3}$

correct

Virtulian Bird
Watermarkly

A closed disc with center z_0 and radius ε is defined by _____

Choices:

$\{z \in \mathbb{C} : |z - z_0| \geq \varepsilon\}$

$\{z \in \mathbb{C} : |z + z_0| \geq \varepsilon\}$

$\{z \in \mathbb{C} : |z - z_0| \leq \varepsilon\}$

correct

$\{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$ ✓

$\{z \in \mathbb{C} : |z + z_0| \leq \varepsilon\}$

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Question: 26 (Marks: 1)

f z lies inside the unit circle, then the image of z under the reciprocal function lies ____

Choices:

- on the unit circle
- outside the unit circle
- inside the unit circle
- nowhere in complex plane

Virtulian Bird
Watermarkly

Question: 22 (Marks: 1)

The function $f(z) = \frac{1}{z-3i}$ is discontinuous at $z =$ _____

Choices:

- $3i$ correct
- $-3i$
- 1
- $-i$

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Complex Analysis

Question: 18 (Marks: 1)

_____ was first to introduce notation of $i=\sqrt{-1}$.

Choices:

- Cardano
- Euler
- Rafael Bombelli
- Pythagoras

L.Euler introduce this notation and Cardano introduce complex number

correct

Virtulian Bird

Watermarkly

Question: 17 (Marks: 1)

A closed disc with center z_0 and radius ϵ is defined by _____

Choices:

- $\{z \in \mathbb{C} : |z - z_0| \geq \epsilon\}$
- $\{z \in \mathbb{C} : |z + z_0| \geq \epsilon\}$
- $\{z \in \mathbb{C} : |z - z_0| \leq \epsilon\}$ correct
- $\{z \in \mathbb{C} : |z + z_0| \leq \epsilon\}$

Virtulian Bird
Watermarkly

Complex Analysis and Differential Geometry

Question: 26 (Marks: 1)

If z lies inside the unit circle, then the image of z under the reciprocal function lies _____

Choices:

- on the unit circle
- outside the unit circle
- inside the unit circle
- nowhere in complex plane

Virtulian Bird

Watermarkly

Complex Analysis and Differential Geometry

Question: 27 (Marks: 1)

$$\lim_{z \to i} \frac{iz^3 - 1}{z + i} = _____$$

Choices:

- 1
- i
- 0
- -1

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Question: 31 (Marks: 5)

If a complex number z satisfies $|z| < 2$, then find upper bound of

$$|z^3 + 3z^2 - 2z + 1|$$

Answer:

Virtulian Bird

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Complex Analysis and Differential Geometry

Question 30 (Marks 5)

Suppose that $f(z_0) = g(z_0) = 0$, $f'(z_0)$ and $g'(z_0)$ exist, where $g'(z_0) \neq 0$. By definition $\frac{dy}{dz} = \lim_{\delta z \to 0} \frac{\delta y}{\delta z}$, show that $\lim_{z \to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$.

Answer:

TIME LEFT
58

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Complex Analysis and Differential Geometry

Question: 23 (Marks: 1)

$\frac{d}{dz}(20z^5) =$ 20*5(z^5-1)
=100(z^4)

Choices:

- $10z^4$
- $20z^5$
- $50z^4$
- $100z^4$

correct ✓

Virtulian Bird
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Complex Analysis and Differential Geometry

Question: 22 (Marks: 1)

The function $f(z)=\frac{1}{z-3i}$ is discontinuous at $z=$ _____

if z=3i then
f(z)=1/3i-3i=1/0
= infinity

Choices:

- 3i
- correct
- -3i
- i
- -i

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Complex Analysis and Differential Geometry

Question: 21 (Marks: 1)

The function $f(z) = \frac{1}{3z+i}$ is discontinuous at $z =$ _____

z=-i/3 then
f(z)=1/3*-i/3 +i
= 1/-i+i = 1/0=
infinity

Choices:

- 3i
- −3i
- $\frac{i}{3}$
- $-\frac{i}{3}$ correct

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Complex Analysis and Differential Geometry

Question: 25 (Marks: 1)

Which of the following are important part(s) of a function?

Choices:

- domain and range
- domain and rule
- range and rule
- rule only

correct

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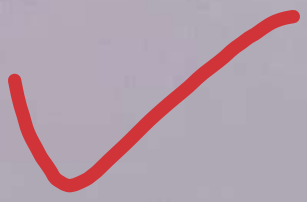
Complex Analysis and Differential Geometry

Questio 1 (Marks: 1)

If $z_1 = 2+i$, $z_2 = 1+3i$, then $i\,\mathrm{Re}(z_1 - z_2)$ = _____.

Choice

1



i

2i

2

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Question 2 (Marks: 1)

The argument of the number, $-1+i$ is _____.

Choice

45°



90°

135°

180°

Virtulian Bird

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Questio 3 (Marks: 1)

$$f(z)=\frac{1}{z^2+4}$$

The function is defined everywhere in the finite plane except at the points _____.

Choice

- $z=\pm i$
- $z=\pm 2$
- $z=\pm 2i$
- $z=\pm 1$

Correct Answer.

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Questio 5 (Marks: 1)

$$\lim_{z \to \infty} \frac{z^2+1}{z-1} =$$

Choice

- π
- i
- $-i$
- ∞

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Question 4 (Marks: 1)

A function of a complex variable is said to be continuous in a region C if it is continuous at _____.

Choice

- each point in C
- one point in C
- two points in C
- nowhere in C

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Complex Analysis and Differential Geometry

Questio 4 (Marks: 1)

A function of a complex variable is said to be continuous in a region C if it is continuous at _____.

Choice

each point in C



one point in C

two points in C

nowhere in C

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Question 8 (Marks: 1)

Let $f(z)=\frac{1}{z}$ is discontinuous at $z=$ _____.

Choice

0



1

-1

i

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Watermarkly

Complex Analysis and Differential Geometry

Questio 7 (Marks: 1)

Square of $4-i$ is _____.

Choice

- $4-i$
- $15-8i$
- $8-2i$
- $8+2i$

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Complex Analysis and Differential Geometry

Question 6 (Marks: 1)

The magnitude of $\exp(-2+3i)$ is _____.

Choice

- $\exp(-2*3)$
- $\exp(2*3)$
- $\exp(3)$
- $\exp(-2)$ ✓

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Complex Analysis and Differential Geometry

Questio 9 (Marks: 1)

The Cauchy-Riemann equations on a pair of real-valued functions of two real variables $u(x, y)$ and $v(x, y)$ are _____.

Choice

$U_x = V_y$ and $U_y = -V_x$

$U_x = V_x$ and $U_y = -V_y$

$U_y = V_y$ and $U_x = -V_x$

$U_x = -V_y$ and $U_y = -V_x$

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Complex Analysis and Differential Geometry

Questio 10 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $V_z =$ _____

Choice

-1

1



0

2

Virtulian Bird

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Question 11 (Marks: 1)

The Polar Cauchy–Riemann equations on a pair of real-valued functions of two real variables u and v are _____.

Choice

- $rU_r = V_\theta$ and $U_\theta = -rV_r$
- $U_r = V_\theta$ and $U_\theta = -V_r$
- $rU_\theta = V_r$ and $U_r = -rV_\theta$
- $U_r = V_\theta$ and $U_\theta = V_r$

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Complex Analysis and Differential Geometry

Questio 14 (Marks: 1)

The limit of $f(z)=\frac{1}{z^2+4}$ at $z=1$ is _____.

Choice

4

1/3

1/4

1/5

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Questio 13 (Marks: 1)

The limit of $f(z)=5z-10$ at $z=i+1$ is _____.

Choice

- $3i-3$
- $5i-5$
- $5i-3$
- $3i-5$



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Questio 12 (Marks: 1)

$$f(z)=\frac{1}{re^{i\theta}}$$

n the complex valued function , the value of $U_r =$ _____

Choice

- $\frac{1}{r}$
- $\frac{\cos\theta}{r}$
- $\cos\theta$
- $e^{i\theta}$

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Complex Analysis and Differential Geometry

Questio 16 (Marks: 1)

Let $f(z)=z^3-3z+12$, then $f'(z)$ is _____.

Choice

- z^2-3
- $3z^2-3$
- z^2-1
- z^3-3

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Complex Analysis and Differential Geometry

Questio 15 (Marks: 1)

Let $f(z)=3z^2+5z+10$, then $f'(z)$ is _____.

Choice

- $5z$
- $6z$
- $6z-5$
- $6z+5$

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Questio 15 (Marks: 1)

Let $f(z)=3z^2+5z+10$, then $f'(z)$ is _____.

Choice

- $5z$
- $6z$
- $6z-5$
- $6z+5$

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Question 18 (Marks: 1)

Let $f(z)=2z^2-7z+1$, then $f'(-2i)$ is _____.

Choice

-2i+1

-4i-7

-8i-7

-8i+1

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Complex Analysis and Differential Geometry

Questio 17 (Marks: 1)

Let $f(z)=2z^3+2z-7$, then $f'(z)$ is _____.

Choice

$6z^2+2$



z^2+2

$2z^2-7$

$2z^3+2$

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Complex Analysis and Differential Geometry

Questio 19 (Marks: 1)

Which of the following statement is TRUE?

Choice

- The set R of real numbers is an extension of set C of complex numbers.
- The set C of complex numbers is an extension of set R of real numbers.
- The set Q of rational numbers is an extension of set Q' of irrational numbers.
- The set Q' of irrational numbers is an extension of set Z of integers

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Complex Analysis and Differential Geometry

Questio 20 (Marks: 1)

The complex number $z = x + iy$ is real if and only if _____.

Choice

- $x = 0$
- $y = 0$
- Both x and y must be 0.
- y must not be 0.

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Questio 21 (Marks: 1)

If $|z| = |3 + i4| = 5$, then $|\bar{z}| =$ _____.

Choice

3

4

5

6

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Questio 21 (Marks: 1)

If $|z| = |3 + i4| = 5$, then $|\bar{z}| =$ _____.

Choice

3

4

5

6

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Questio 24 (Marks: 1)

The function $f(z)=\frac{1}{z^2+25}$ is discontinuous at $z=$ _____.

Choice

- $\pm 2i$
- $\pm 3i$
- $\pm 4i$
- $\pm 5i$

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Complex Analysis and Differential Geometry

Questio 23 (Marks: 1)

Let $z_1 = r_1 e^{\theta_1}$ be a point inside the unit circle, then the reciprocal $\frac{1}{z_1} = \frac{1}{r_1 e^{\theta_1}}$ will lie _____.

Choice

- inside the unit circle.
- outside the unit circle.
- on the unit circle.
- nowhere in complex plane.

Virtulian Bird

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Complex Analysis and Differential Geometry

Questio 22 (Marks: 1)

The function $f(z) = \dfrac{1}{2z - i}$ is discontinuous at $z =$ _____.

Choice

- i
- $-i$
- $\frac{i}{2}$
- $-\frac{i}{2}$

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Complex Analysis and Differential Geometry

Questio 27 (Marks: 1)

The punctured open disc with radius ε and center at z_0 is defined by _____

Choice

- $\{z \in C : |z - z_0| < \varepsilon\}$
- $\{z \in C : |z - z_0| \geq \varepsilon\}$
- $\{z \in C : 0 < |z - z_0| < \varepsilon\}$
- $\{z \in C : |z - z_0| > \varepsilon\}$

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Complex Analysis and Differential Geometry

Questio 26 (Marks: 1)

The expression $(\cos\theta + i\sin\theta)^3$ can be written as _____.

Choice

$3\cos\theta + i\,3\sin\theta$

$\cos 3\theta + i\sin 3\theta$

$3\cos\theta + i\sin\theta$

$3\cos 3\theta + i\,3\sin 3\theta$

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Questio 25 (Marks: 1)

$f(z) = \frac{1}{z} = z^{-1}$ is a polynomial.

Choice

True

False

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Watermarkly

Questio 27 (Marks: 1)

The punctured open disc with radius ε and center at z_0 is defined by ____

Choice

- $\{z \in C : |z - z_0| < \varepsilon\}$
- $\{z \in C : |z - z_0| \geq \varepsilon\}$
- $\{z \in C : 0 < |z - z_0| < \varepsilon\}$ ✓
- $\{z \in C : |z - z_0| > \varepsilon\}$

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Questio 28 (Marks: 3)

Evaluate $\lim_{z \to \infty} \left(\frac{5z^2+1}{3z^2-z} \right)$

Answe

Virtulian Bird

Watermarkly

Question 29 (Marks: 3)

Let $S : \{z-3\} \leq 5$ be a region. Find its interior points, exterior points and boundary points.

Answ

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Questio 30 (Marks: 5)

Find $g'(w)$, where $g(w) = \dfrac{(3+i)w^2}{2w-1}$.

Answe

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Complex Analysis and Differential Geometry

Questio 31 (Marks: 5)

$$f(z)=\frac{z^3}{(z+i)^3} \text{ at } z=i$$

Find the derivative of

Answe

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